

Test case generator Langflow

User story:

User Story: HC-310 — Doctor Prescription Verification Workflow

Summary:

As a doctor, I want an automated prescription verification system so that I can review, approve, or modify prescriptions digitally before they are sent to the pharmacy.

Description:

The system should enable doctors to view medication details prescribed by other physicians or generated automatically from patient diagnosis. It should allow verification of dosage, potential interactions, and duplicate drugs. Once verified, prescriptions should automatically sync with the pharmacy module for patient dispensing.

Acceptance Criteria:

Given a doctor with access to a patient record,

When they open the prescription module,

Then all pending prescriptions should be displayed for verification.

Given a prescription with errors (dosage conflict or duplicate medicine),

When the doctor reviews it,

Then the system should highlight issues and provide AI-based recommendations for correction.

Given a verified prescription,

When it is approved,

Then it should automatically update the pharmacy dispensing queue.

Prompt template :

I - Instruction:

You are a Senior QA Engineer with expertise in healthcare systems. Your task is to generate comprehensive test cases for the given user story. Ensure coverage for functional, negative, edge, boundary, and integration scenarios, with special focus on patient safety and data accuracy.

You must cover:

- Positive scenarios
- Negative scenarios
- Validation scenarios
- Boundary conditions
- Error and failure handling

C - Context:

The system is a Doctor Prescription Verification Workflow. It allows doctors to:

- View prescriptions (manual or auto-generated)
- Verify dosage accuracy
- Detect duplicate medicines
- Identify drug interactions
- Receive AI-based recommendations for corrections
- Approve prescriptions
- Sync verified prescriptions to pharmacy dispensing system

The system involves:

- UI (Prescription module, highlighting errors)
- Backend validation (dosage checks, duplicate detection)
- AI recommendations (suggest corrections)

- Integration (pharmacy module sync)

Acceptance Criteria to validate:

1. Display all pending prescriptions for verification
2. Highlight errors (dosage conflicts, duplicate medicines)
3. Provide AI-based recommendations
4. Sync approved prescriptions to pharmacy queue

E - Examples:(Reference Format Only)

Test Case ID: TC_001

Title: Verify system detects duplicate medicines

Module: Prescription Verification

Priority: High

Preconditions:

- Doctor logged in
- Prescription contains duplicate drug

Steps:

1. Open prescription module
2. Select prescription
3. Review medication list

Test Data:

- Drug A listed twice

Expected Result:

- System highlights duplicate drugs
- Warning message displayed
- AI recommendation shown

P - Persona:

Act as a highly detail-oriented QA expert in healthcare applications. Ensure strict validation for safety-critical scenarios. Think like a tester who prioritizes patient safety, regulatory compliance, and system reliability.

O - Output Format:

Generate multiple test cases with the following structure:

For each test case:

- Test Case ID
- Title
- Module (Prescription / AI Validation / Pharmacy Integration / UI)
- Priority (High / Medium / Low)
- Preconditions
- Test Steps (numbered and clear)
- Test Data (if applicable)
- Expected Result

Coverage Requirements:

- Minimum 15–20 test cases
- Include:
 - Positive scenarios

- Negative scenarios
- Edge cases
- Boundary conditions
- AI recommendation validation
- Integration validation with pharmacy system
- Error handling scenarios

T - Tone:

Professional, precise, and safety-focused. Avoid ambiguity. Ensure clarity and completeness in all test cases.

Output:

Test Cases for Doctor Prescription Verification Workflow

Test Case 1: Display Pending Prescriptions

- **Test Case ID:** TC_001
- **Title:** Verify system displays all pending prescriptions for verification
- **Module:** Prescription Verification
- **Priority:** High
- **Preconditions:** Doctor logged in, patient record accessible
- **Test Steps:**
 1. Log in as a doctor.
 2. Access a patient's record.
 3. Open the prescription module.
- **Test Data:** Patient with pending prescriptions
- **Expected Result:** All pending prescriptions are displayed for verification.

Test Case 2: Highlight Dosage Conflicts

- **Test Case ID:** TC_002
- **Title:** Verify system highlights dosage conflicts
- **Module:** Prescription Verification
- **Priority:** High
- **Preconditions:** Doctor logged in, prescription with dosage conflict
- **Test Steps:**
 1. Log in as a doctor.
 2. Access a patient's record with a prescription containing a dosage conflict.
 3. Review the prescription.
- **Test Data:** Prescription with a known dosage conflict
- **Expected Result:** System highlights the dosage conflict and provides a warning message.

Test Case 3: Detect Duplicate Medicines

- **Test Case ID:** TC_003
- **Title:** Verify system detects duplicate medicines
- **Module:** Prescription Verification
- **Priority:** High
- **Preconditions:** Doctor logged in, prescription with duplicate medicines
- **Test Steps:**
 1. Log in as a doctor.
 2. Access a patient's record with a prescription containing duplicate medicines.
 3. Review the medication list.
- **Test Data:** Prescription with duplicate medicines listed

- **Expected Result:** System highlights duplicate medicines and displays a warning message.

Test Case 4: AI-Based Recommendations

- **Test Case ID:** TC_004
- **Title:** Verify system provides AI-based recommendations for corrections
- **Module:** AI Validation
- **Priority:** High
- **Preconditions:** Doctor logged in, prescription with errors
- **Test Steps:**
 1. Log in as a doctor.
 2. Access a patient's record with a prescription containing errors.
 3. Review the prescription and AI recommendations.
- **Test Data:** Prescription with known errors (dosage conflict or duplicate medicines)
- **Expected Result:** System provides AI-based recommendations for correcting the errors.

Test Case 5: Approve Prescription

- **Test Case ID:** TC_005
- **Title:** Verify approved prescription syncs with pharmacy queue
- **Module:** Pharmacy Integration
- **Priority:** High
- **Preconditions:** Doctor logged in, verified prescription
- **Test Steps:**
 1. Log in as a doctor.
 2. Access a verified prescription.

3. Approve the prescription.

- **Test Data:** Verified prescription
- **Expected Result:** Prescription is automatically updated in the pharmacy dispensing queue.

Test Case 6: Negative Scenario - Invalid Login

- **Test Case ID:** TC_006
- **Title:** Verify system response to invalid doctor login
- **Module:** UI
- **Priority:** Medium
- **Preconditions:** Invalid doctor credentials
- **Test Steps:**
 1. Attempt to log in with invalid credentials.
- **Test Data:** Invalid username or password
- **Expected Result:** System denies access and displays an error message.

Test Case 7: Edge Case - Empty Prescription List

- **Test Case ID:** TC_007
- **Title:** Verify system response to an empty prescription list
- **Module:** Prescription Verification
- **Priority:** Medium
- **Preconditions:** Doctor logged in, patient with no prescriptions
- **Test Steps:**
 1. Log in as a doctor.
 2. Access a patient's record with no prescriptions.
 3. Open the prescription module.

- **Test Data:** Patient with no prescriptions
- **Expected Result:** System displays a message indicating no prescriptions are available for verification.

Test Case 8: Boundary Condition - Maximum Dosage

- **Test Case ID:** TC_008
- **Title:** Verify system response to maximum dosage
- **Module:** Prescription Verification
- **Priority:** Medium
- **Preconditions:** Doctor logged in, prescription with maximum dosage
- **Test Steps:**
 1. Log in as a doctor.
 2. Access a patient's record with a prescription at the maximum dosage.
 3. Review the prescription.
- **Test Data:** Prescription with maximum allowable dosage
- **Expected Result:** System does not highlight the dosage as an error but may provide information on the maximum dosage.

Test Case 9: Error Handling - Network Failure

- **Test Case ID:** TC_009
- **Title:** Verify system response to network failure during prescription sync
- **Module:** Pharmacy Integration
- **Priority:** High
- **Preconditions:** Doctor logged in, verified prescription, simulated network failure
- **Test Steps:**
 1. Log in as a doctor.

2. Access a verified prescription.
 3. Simulate a network failure.
 4. Attempt to approve the prescription.
- ****Test Data:**** Verified prescription, network failure simulation
 - ****Expected Result:**** System displays an error message indicating the failure to sync with the pharmacy queue due to network issues and provides an option to retry.

Test Case 10: Validation Scenario - Drug Interaction

- ****Test Case ID:**** TC_010
- ****Title:**** Verify system detects drug interactions
- ****Module:**** Prescription Verification
- ****Priority:**** High
- ****Preconditions:**** Doctor logged in, prescription with potential drug interactions
- ****Test Steps:****
 1. Log in as a doctor.
 2. Access a patient's record with a prescription containing potential drug interactions.
 3. Review the prescription.
- ****Test Data:**** Prescription with known drug interactions
- ****Expected Result:**** System highlights the potential drug interactions and provides a warning message.

Screenshot of Langflow:

